

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

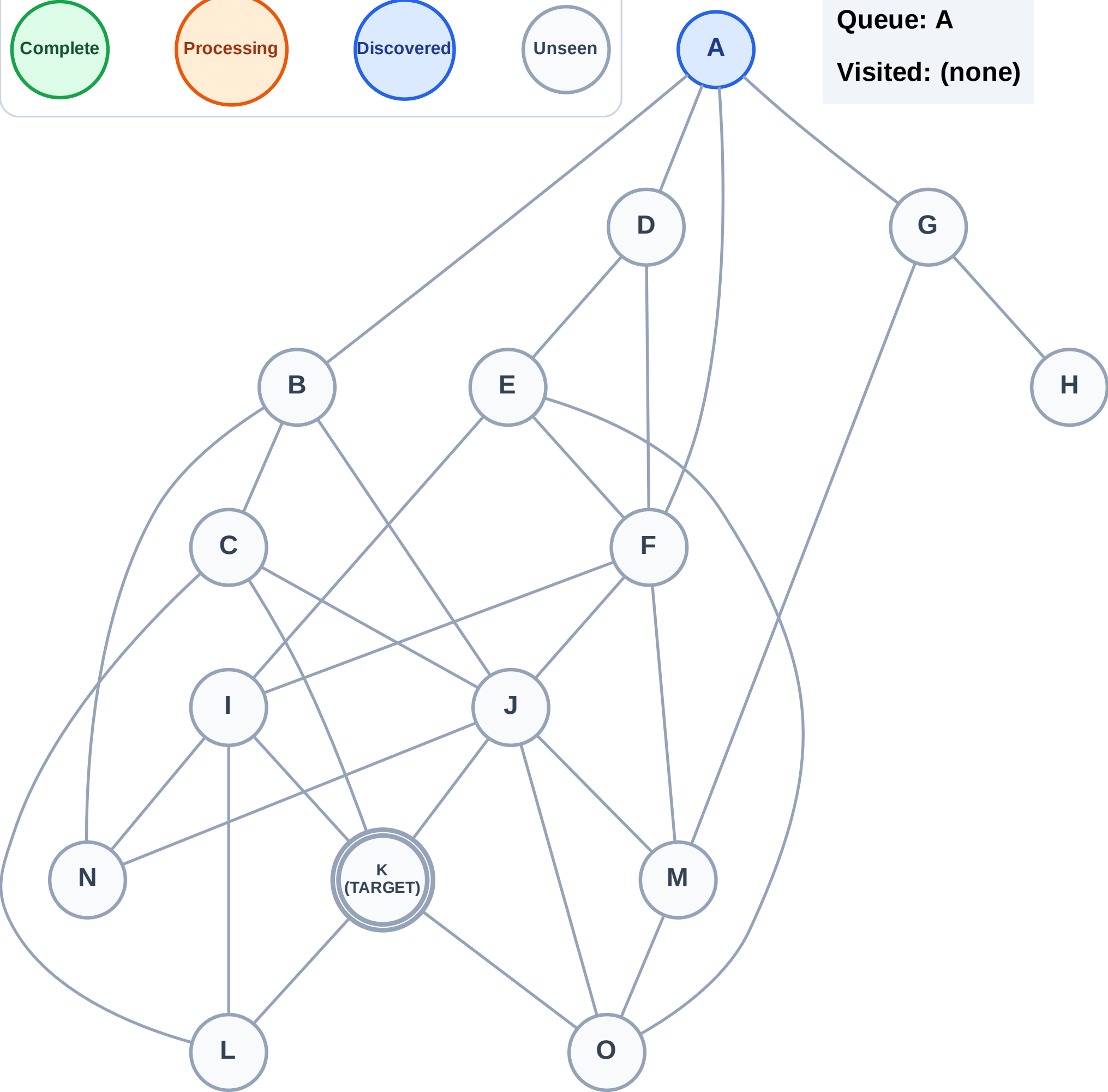
Complete

Processing

Discovered

Unseen

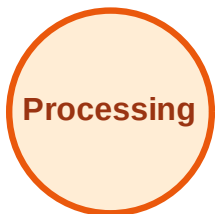
Queue: A
Visited: (none)



BFS Traversal

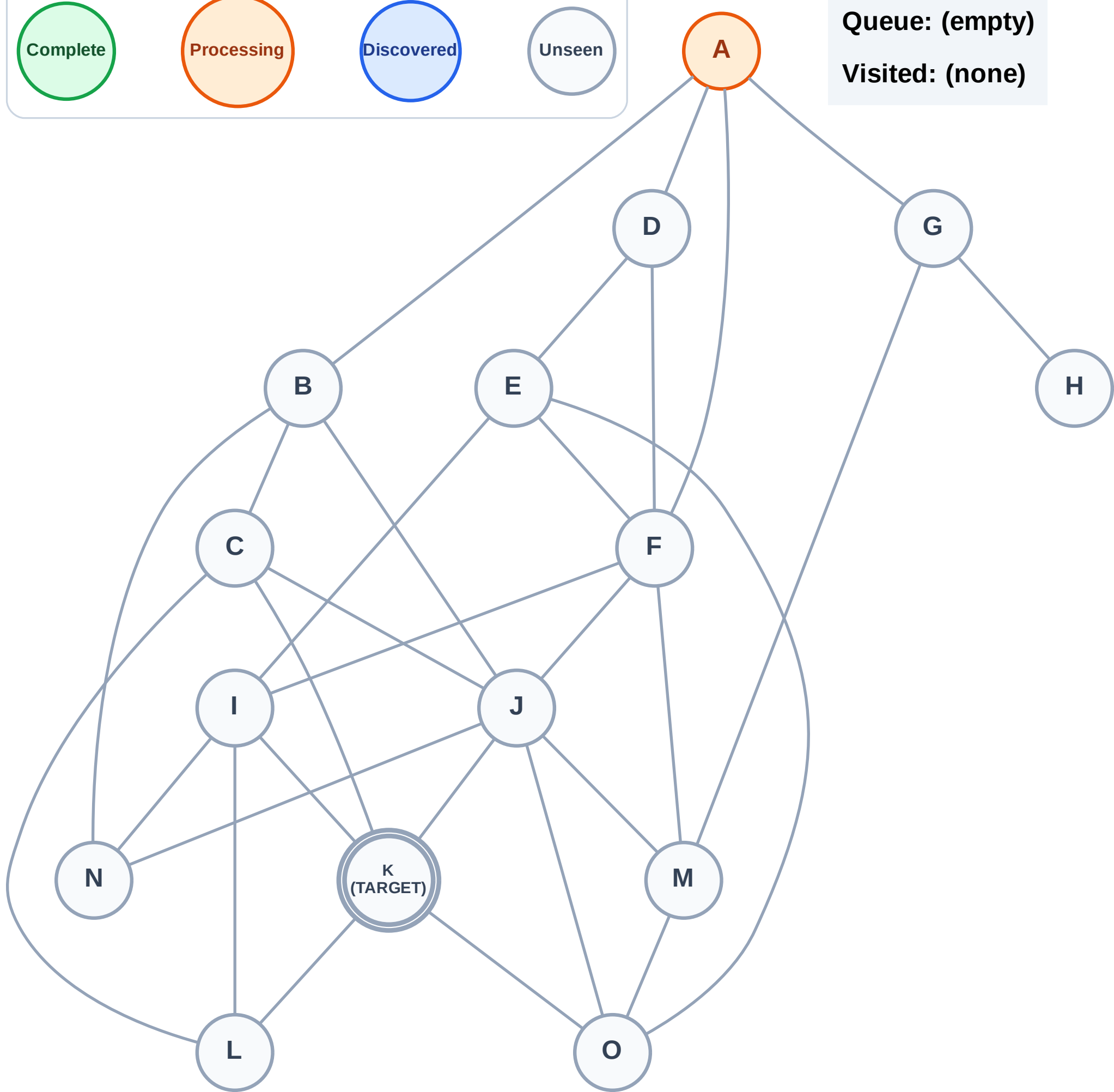
Step 2: Process node A

Legend



Queue: (empty)

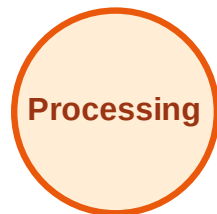
Visited: (none)



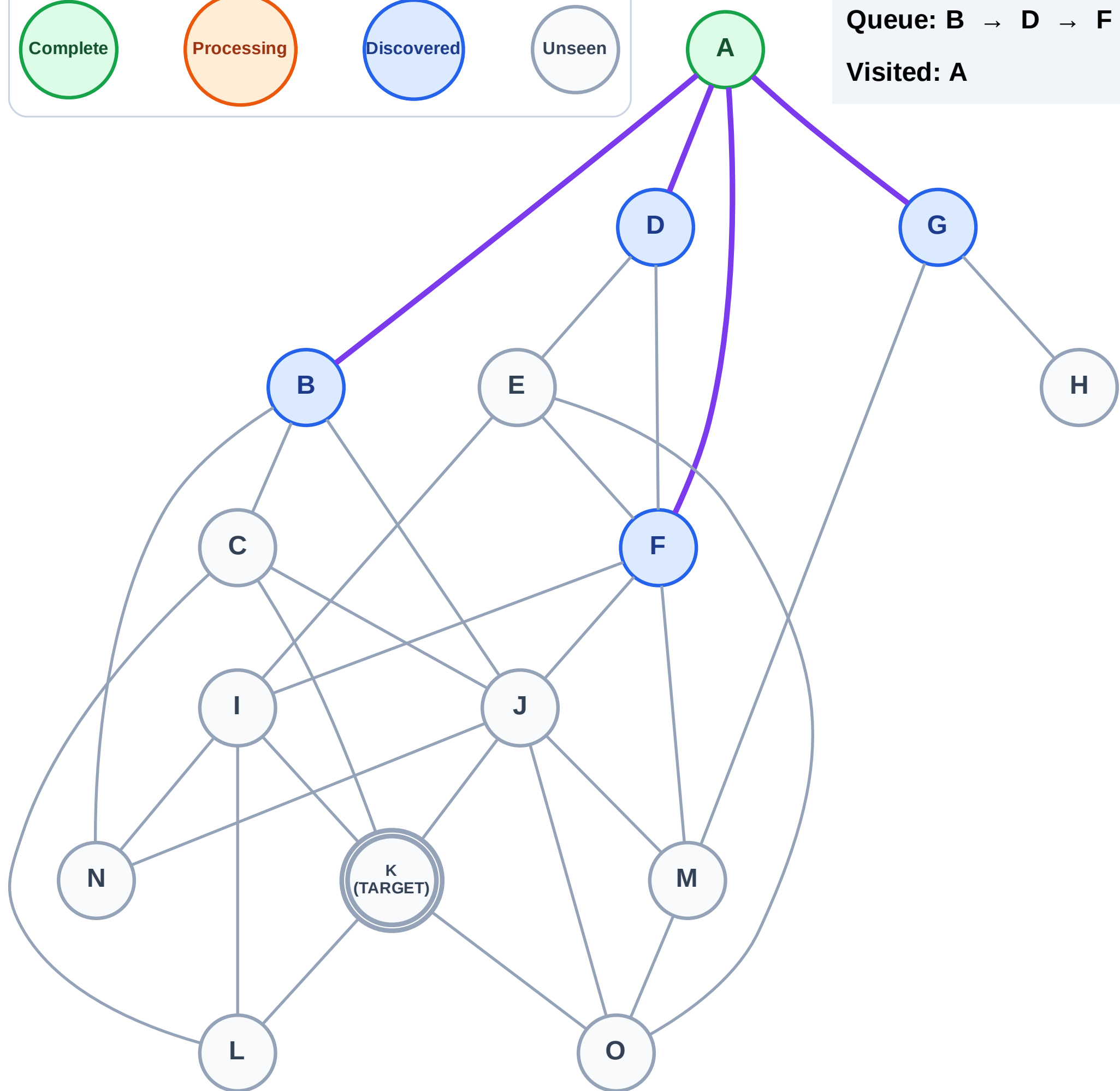
BFS Traversal

Step 3: Discover B, D, F, G from A

Legend



Queue: B → D → F → G
Visited: A



BFS Traversal

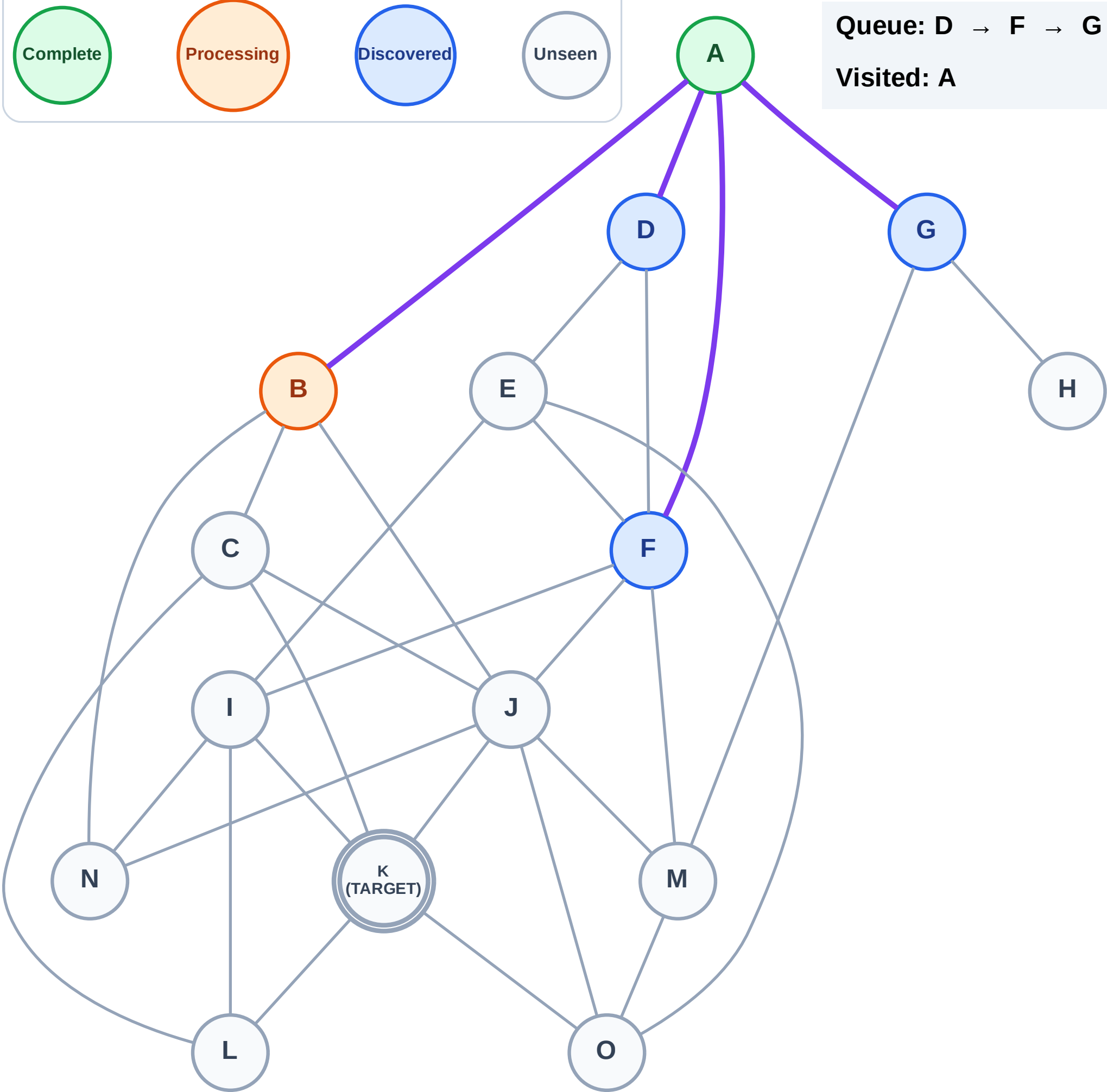
Step 4: Process node B

Legend



Queue: D → F → G

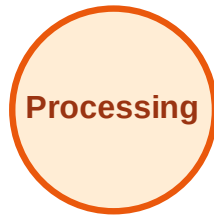
Visited: A



BFS Traversal

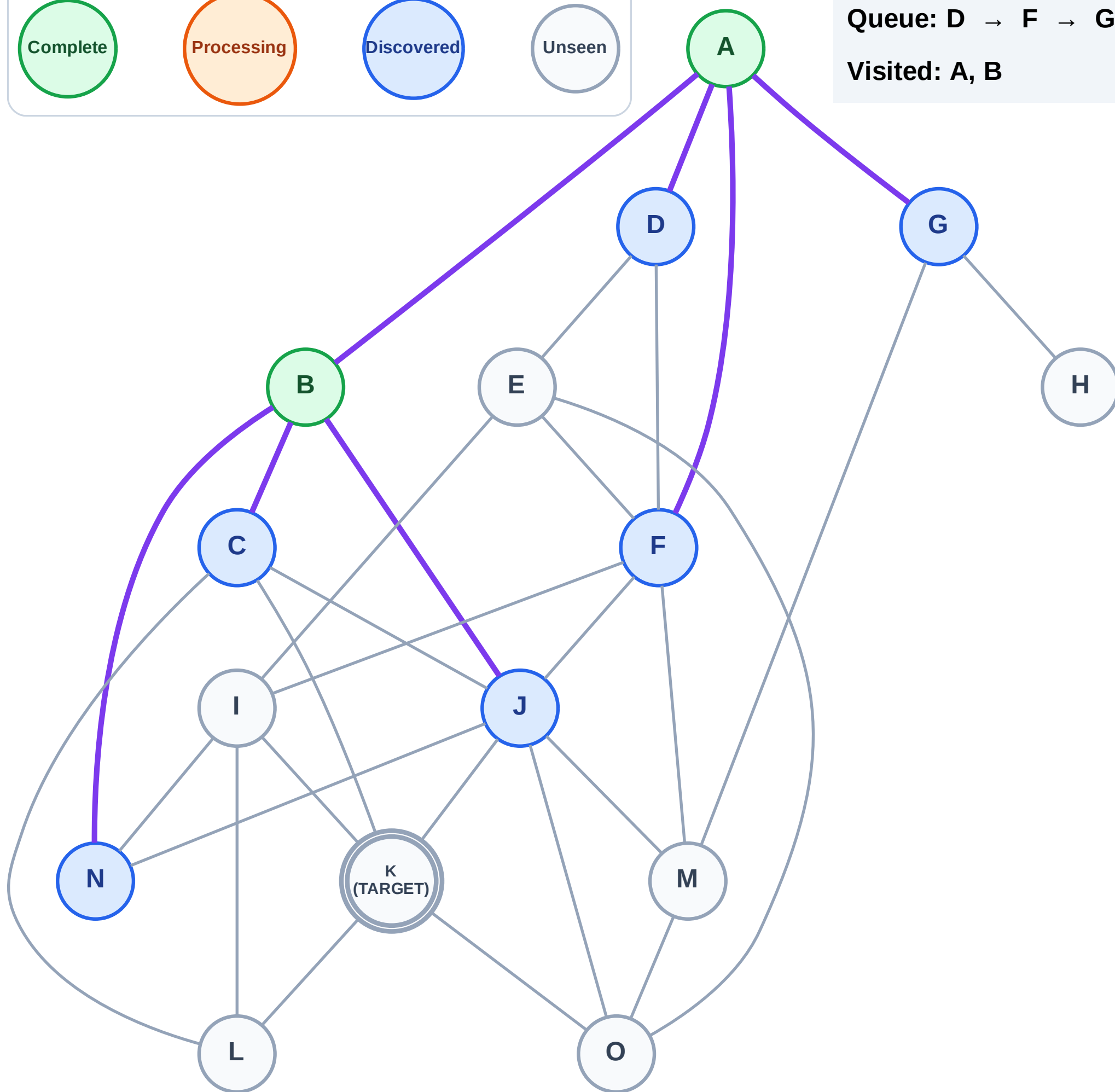
Step 5: Discover C, J, N from B

Legend



Queue: D → F → G → C → J → N

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

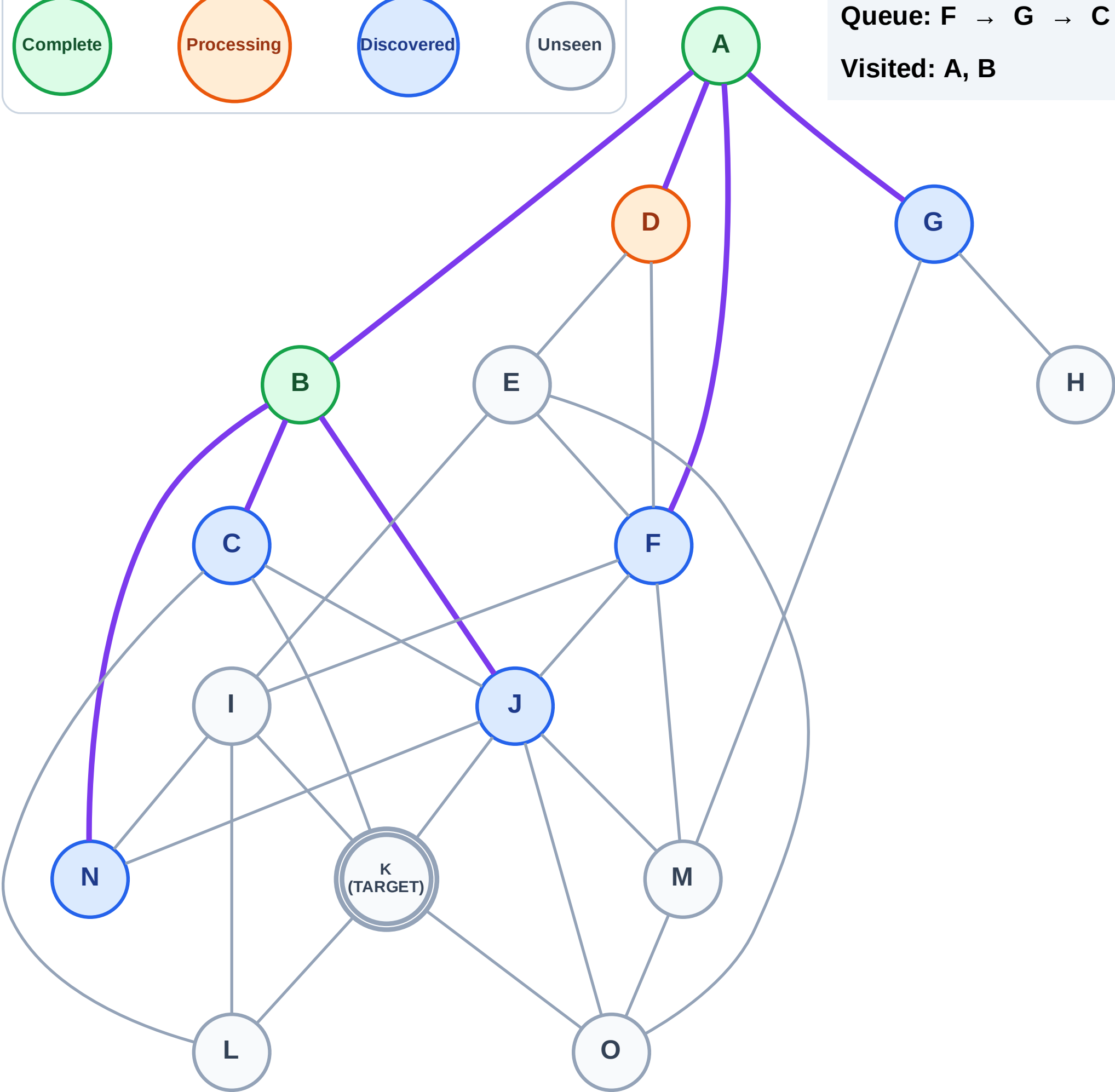
Processing

Discovered

Unseen

Queue: F → G → C → J → N

Visited: A, B



BFS Traversal

Step 7: Discover E from D

Legend

Complete

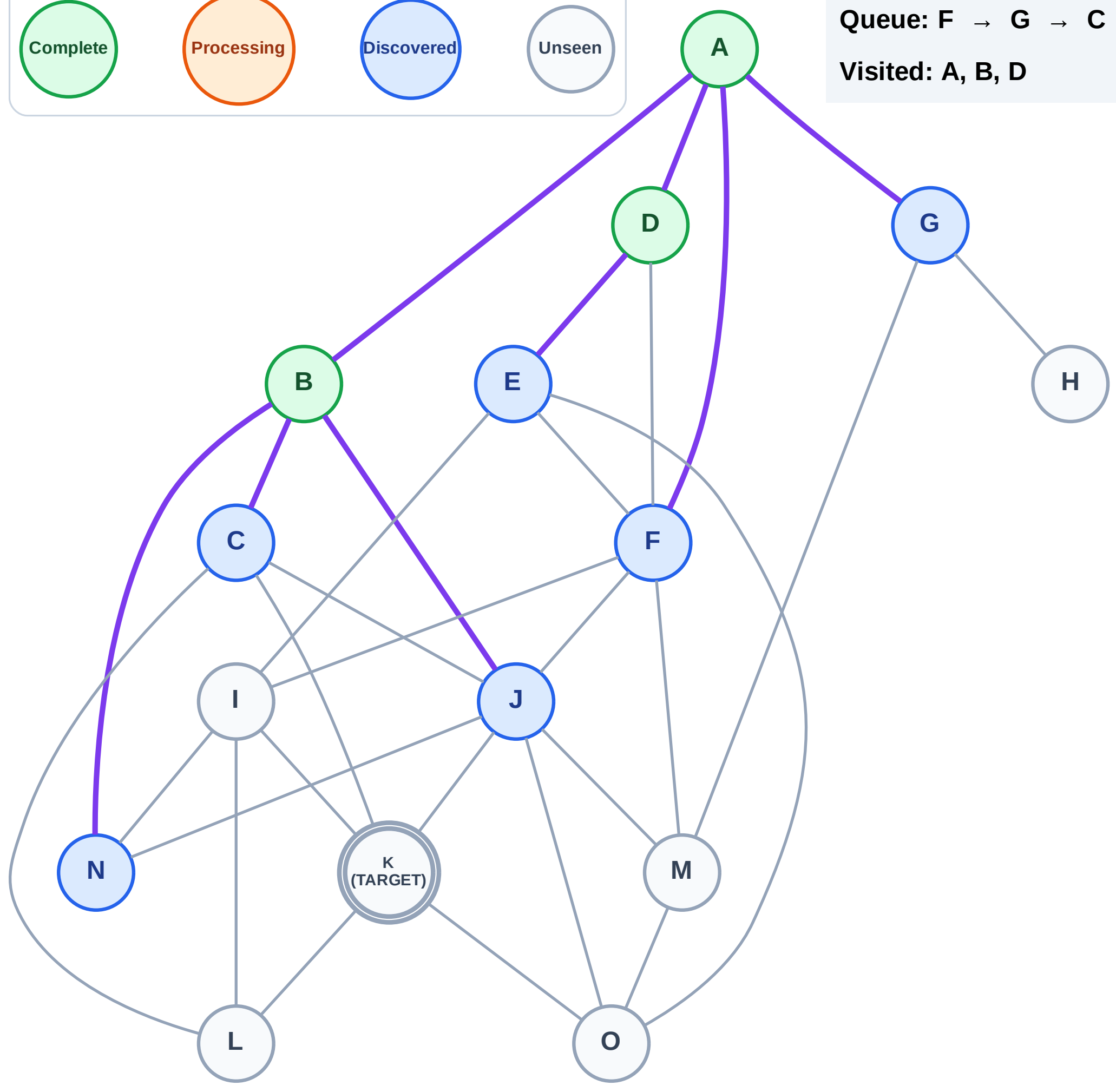
Processing

Discovered

Unseen

Queue: F → G → C → J → N → E

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



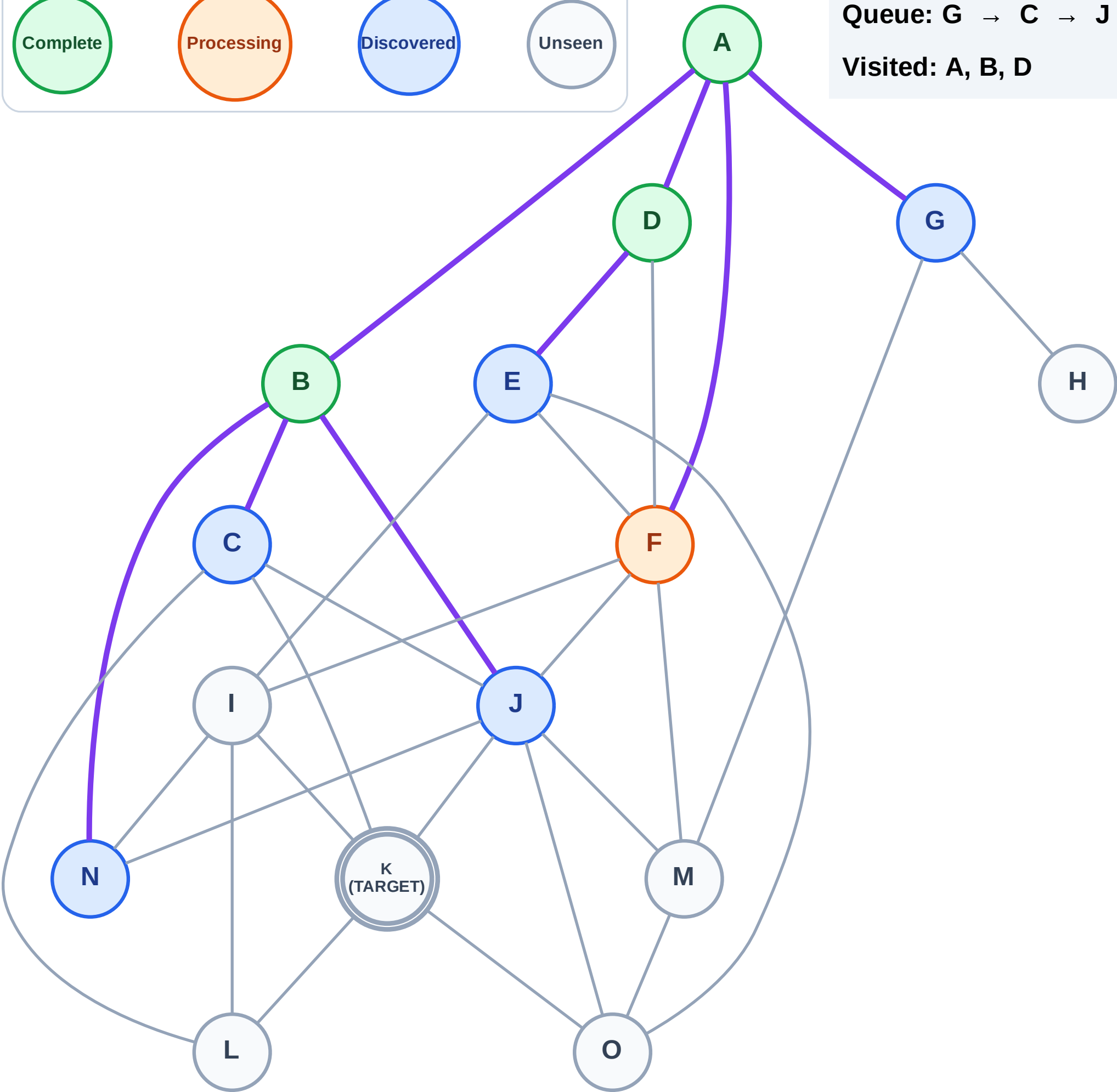
Discovered



Unseen

Queue: G → C → J → N → E

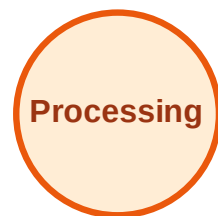
Visited: A, B, D



BFS Traversal

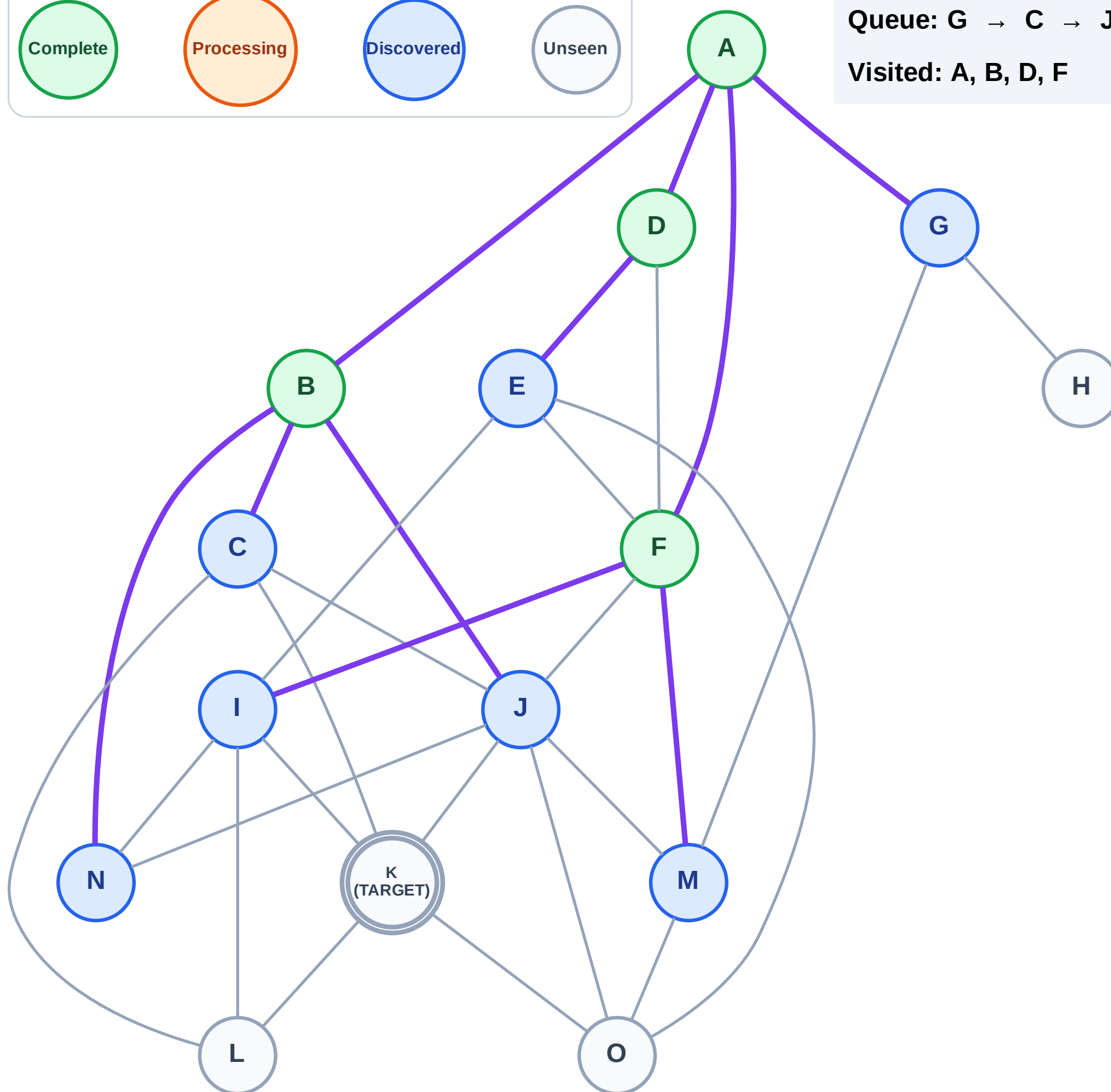
Step 9: Discover I, M from F

Legend



Queue: G → C → J → N → E → I → M

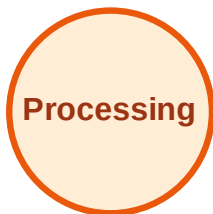
Visited: A, B, D, F



BFS Traversal

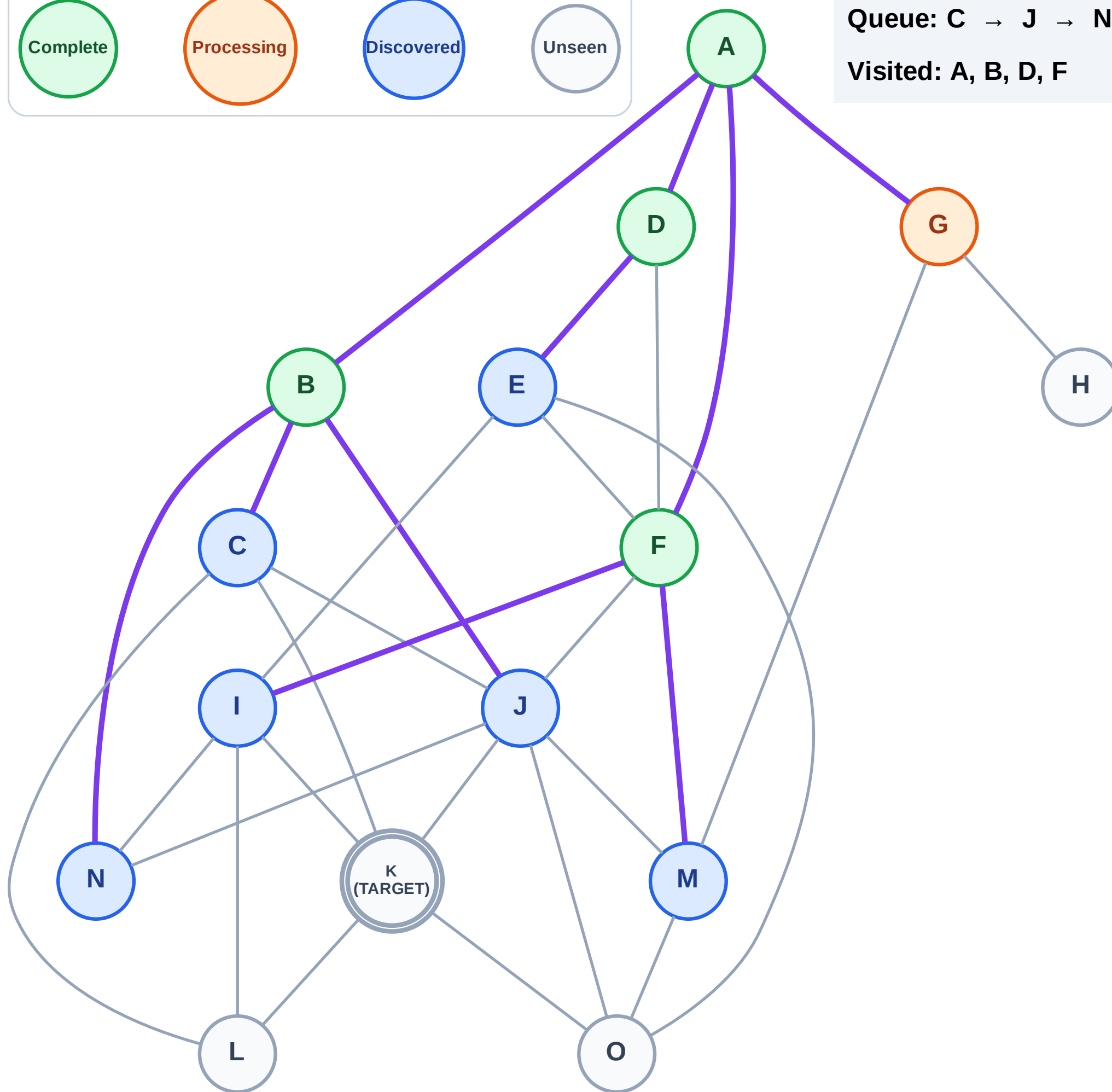
Step 10: Process node G

Legend



Queue: C → J → N → E → I → M

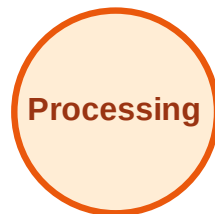
Visited: A, B, D, F



BFS Traversal

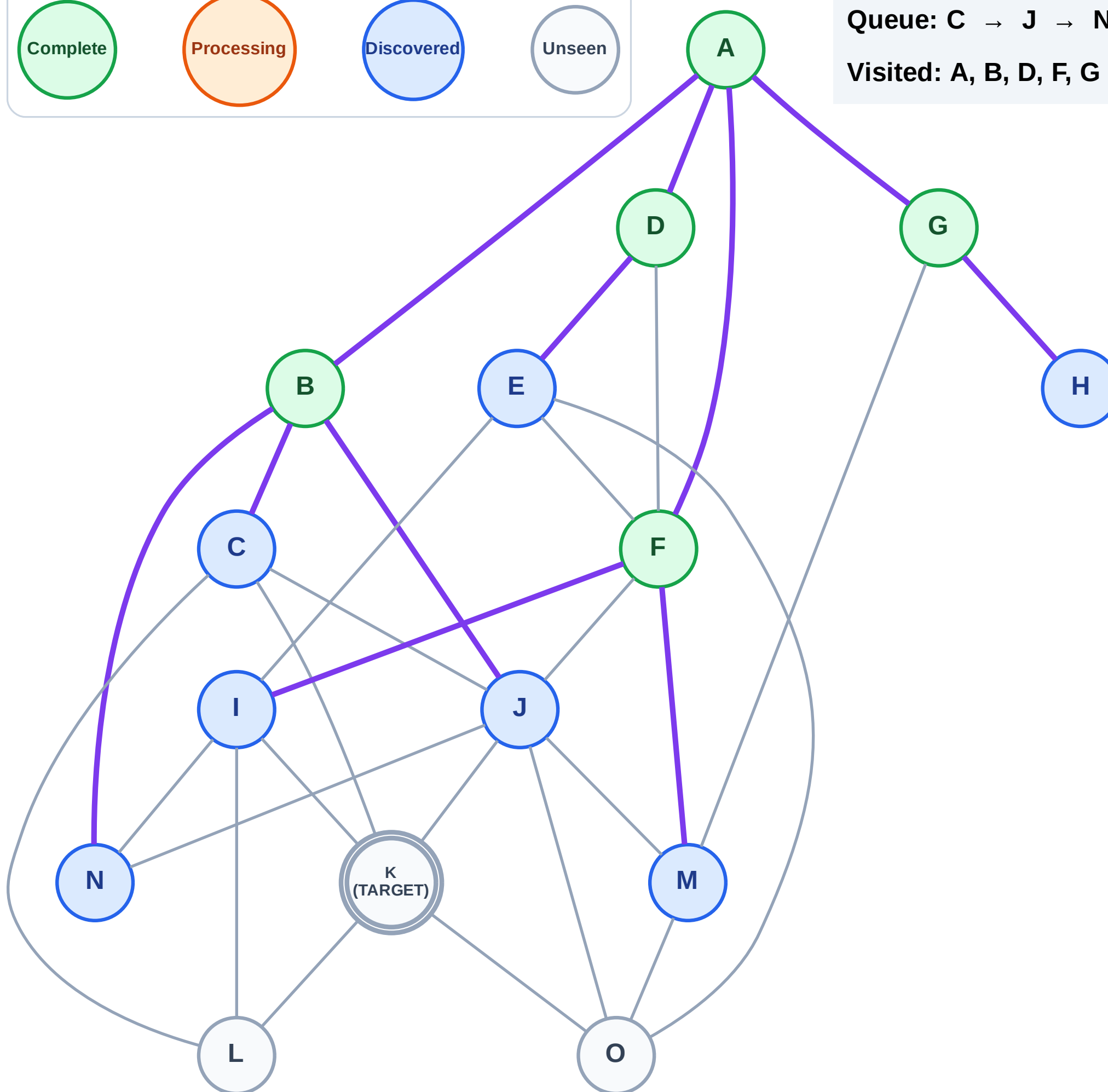
Step 11: Discover H from G

Legend



Queue: C → J → N → E → I → M → H

Visited: A, B, D, F, G



BFS Traversal

Step 12: Process node C

Legend

Complete

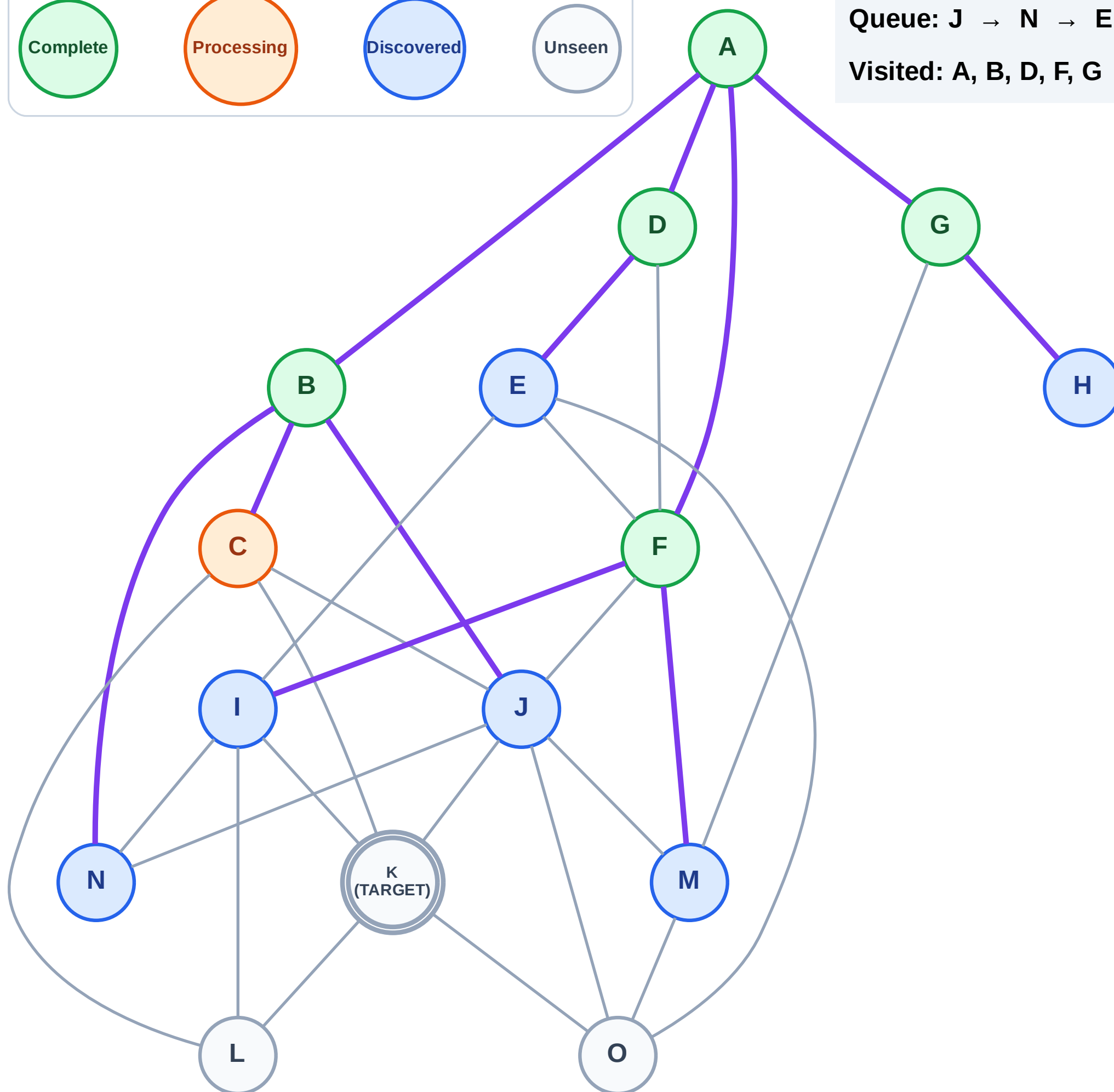
Processing

Discovered

Unseen

Queue: J → N → E → I → M → H

Visited: A, B, D, F, G



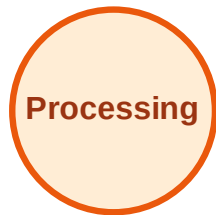
BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Complete



Processing



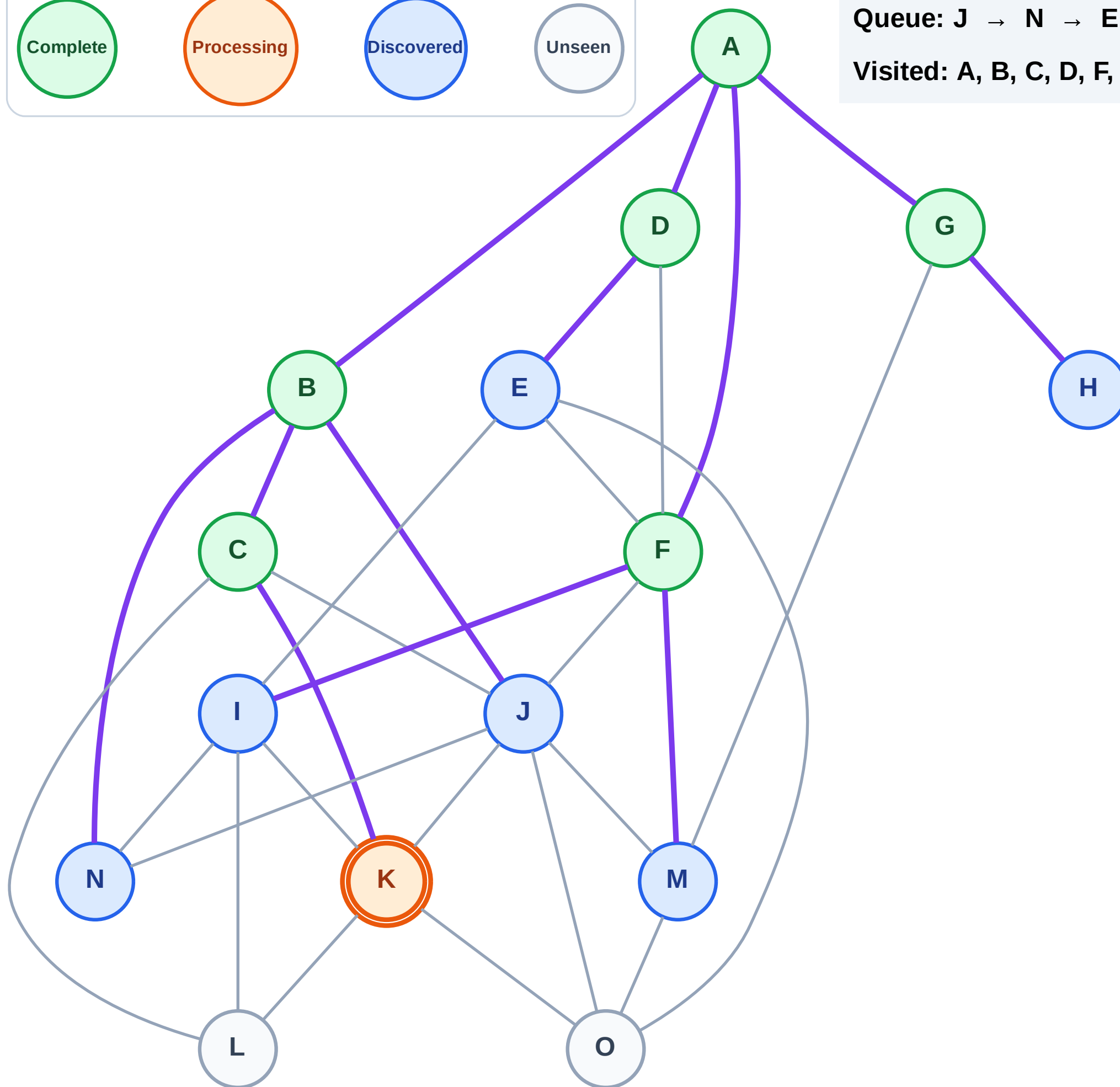
Discovered



Unseen

Queue: J → N → E → I → M → H

Visited: A, B, C, D, F, G, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

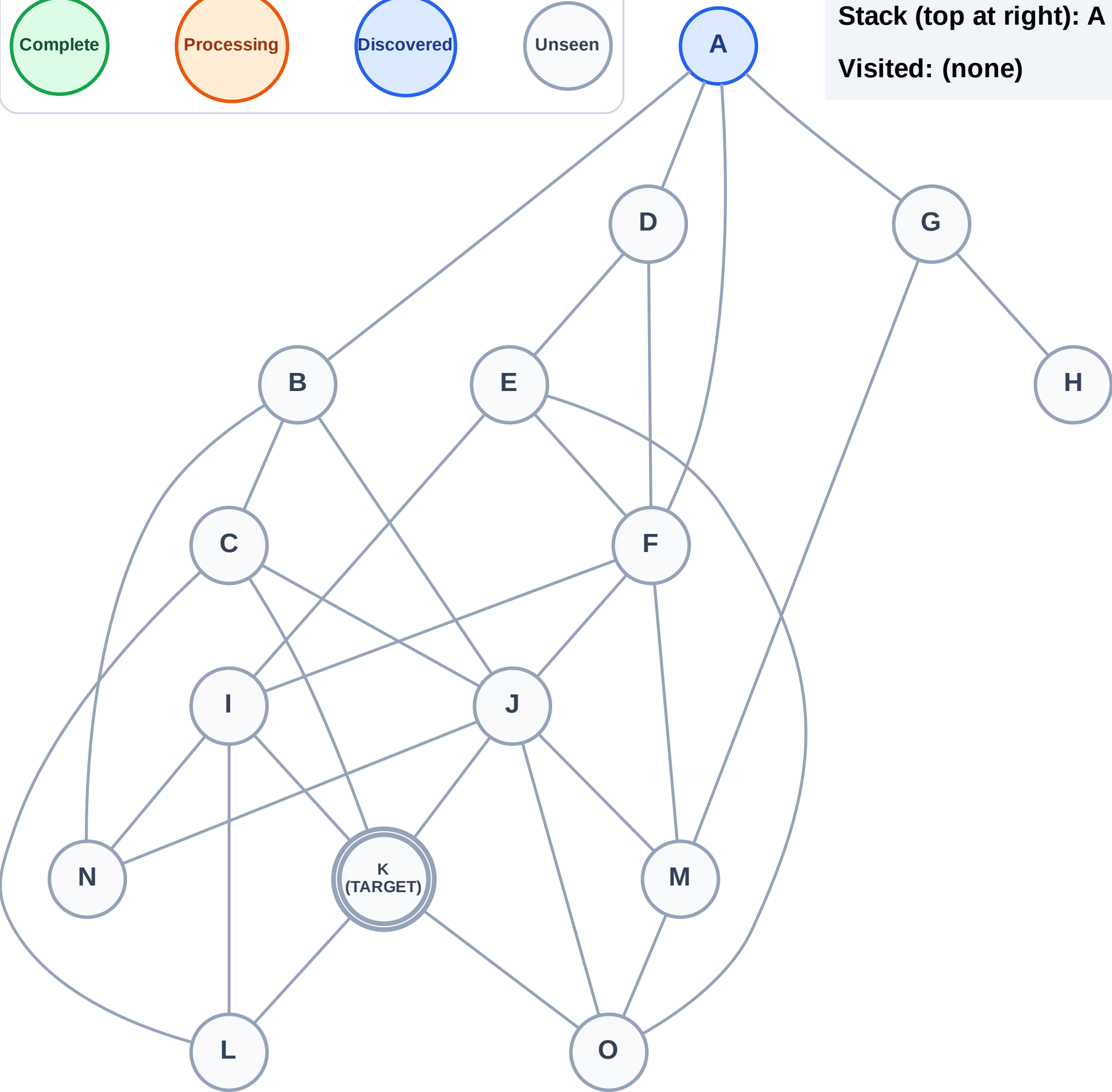
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

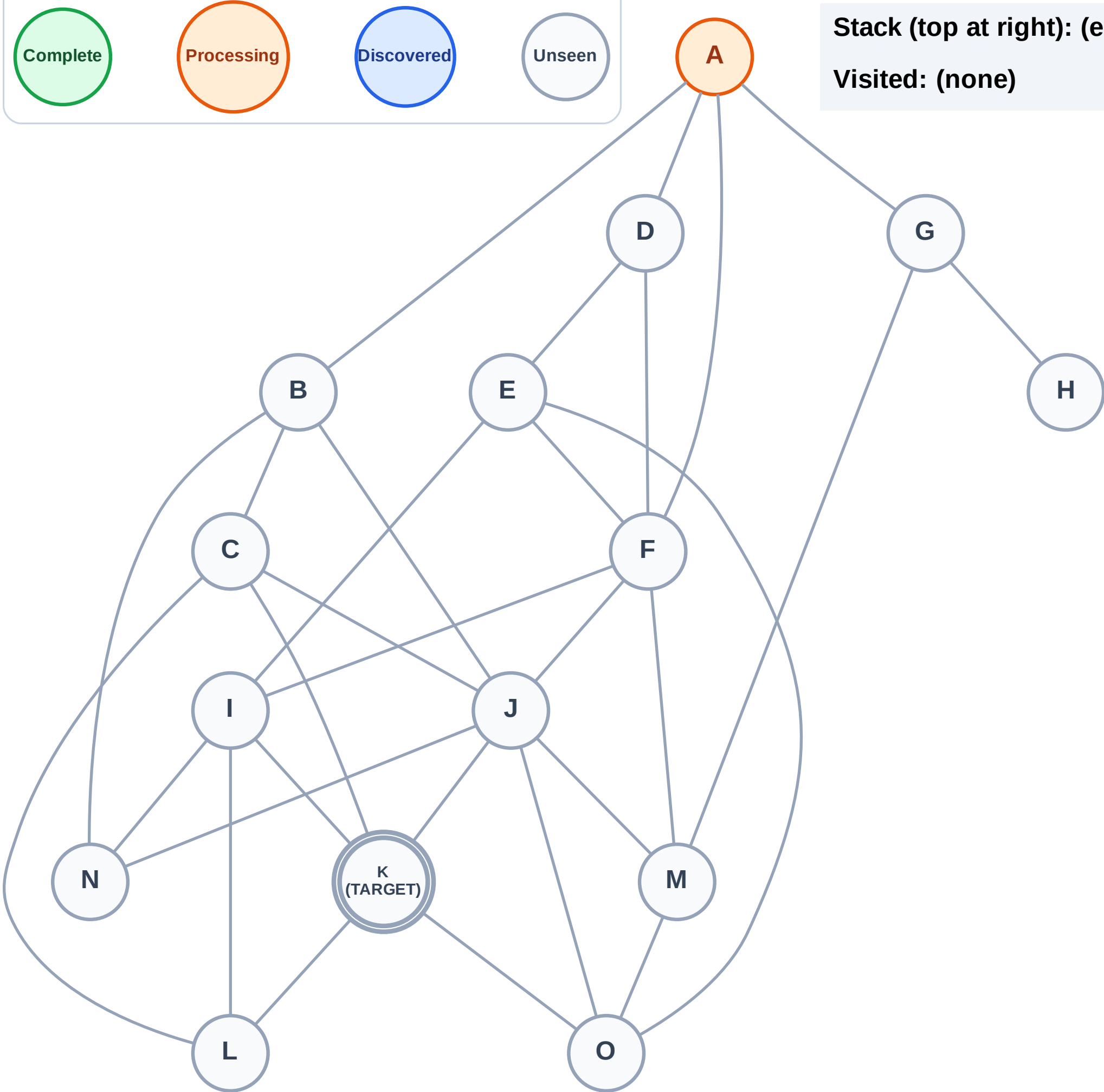
Processing

Discovered

Unseen

Stack (top at right): (empty)

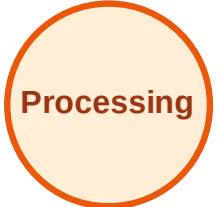
Visited: (none)



DFS Traversal

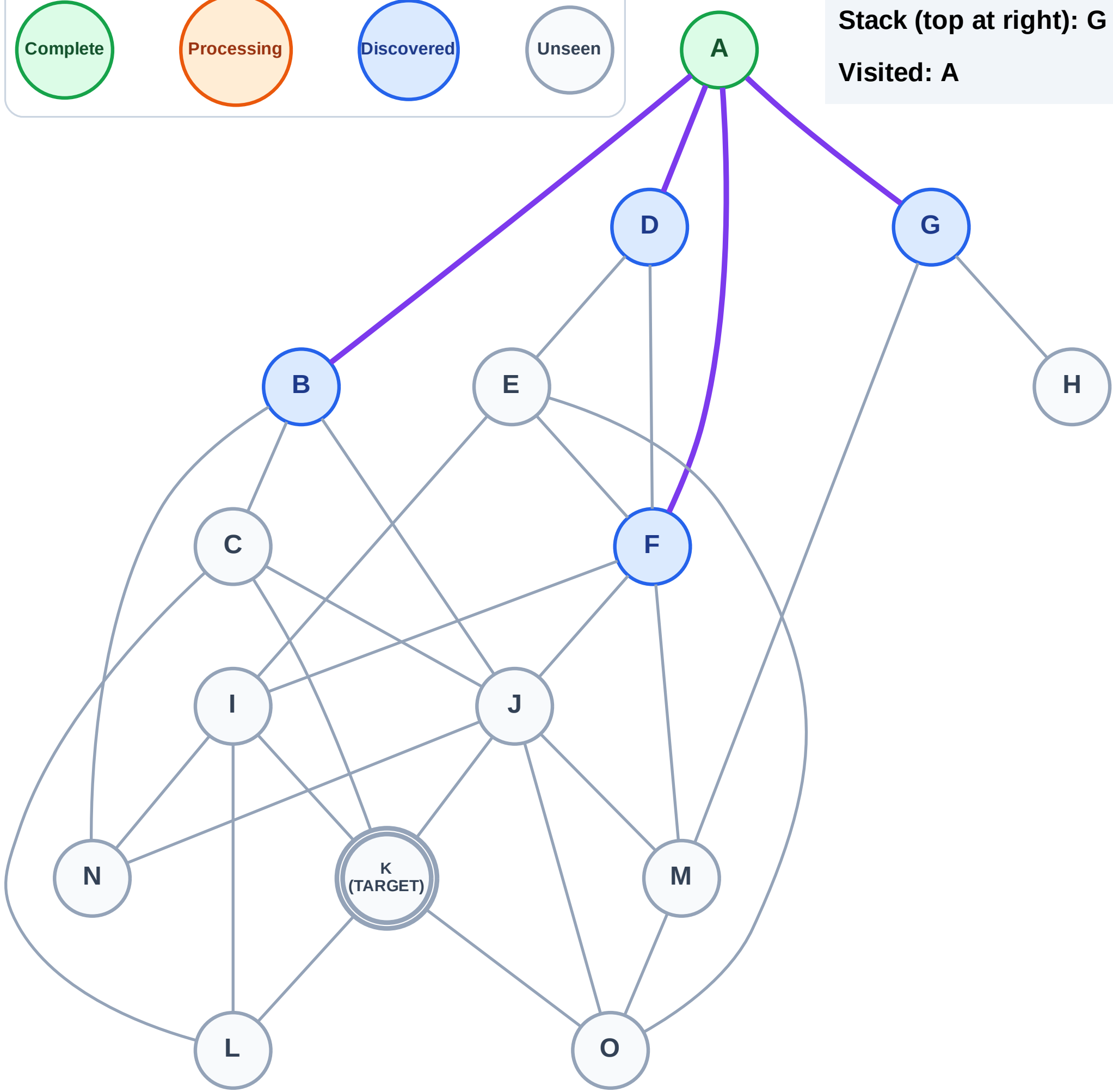
Step 3: Discover G, F, D, B from A

Legend



Stack (top at right): G | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

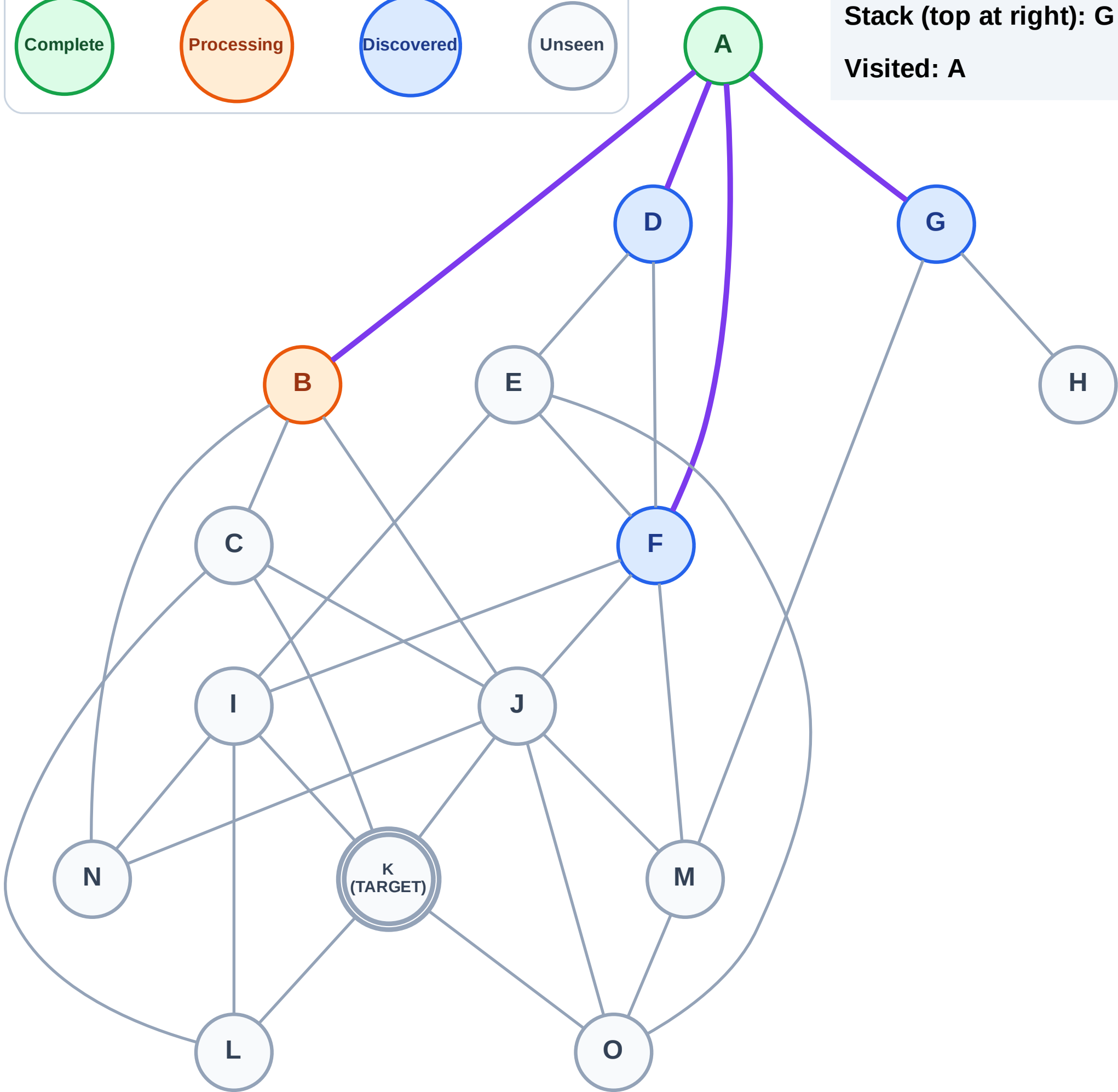
Processing

Discovered

Unseen

Stack (top at right): G | F | D

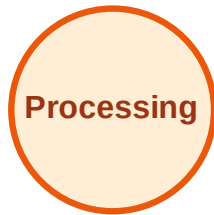
Visited: A



DFS Traversal

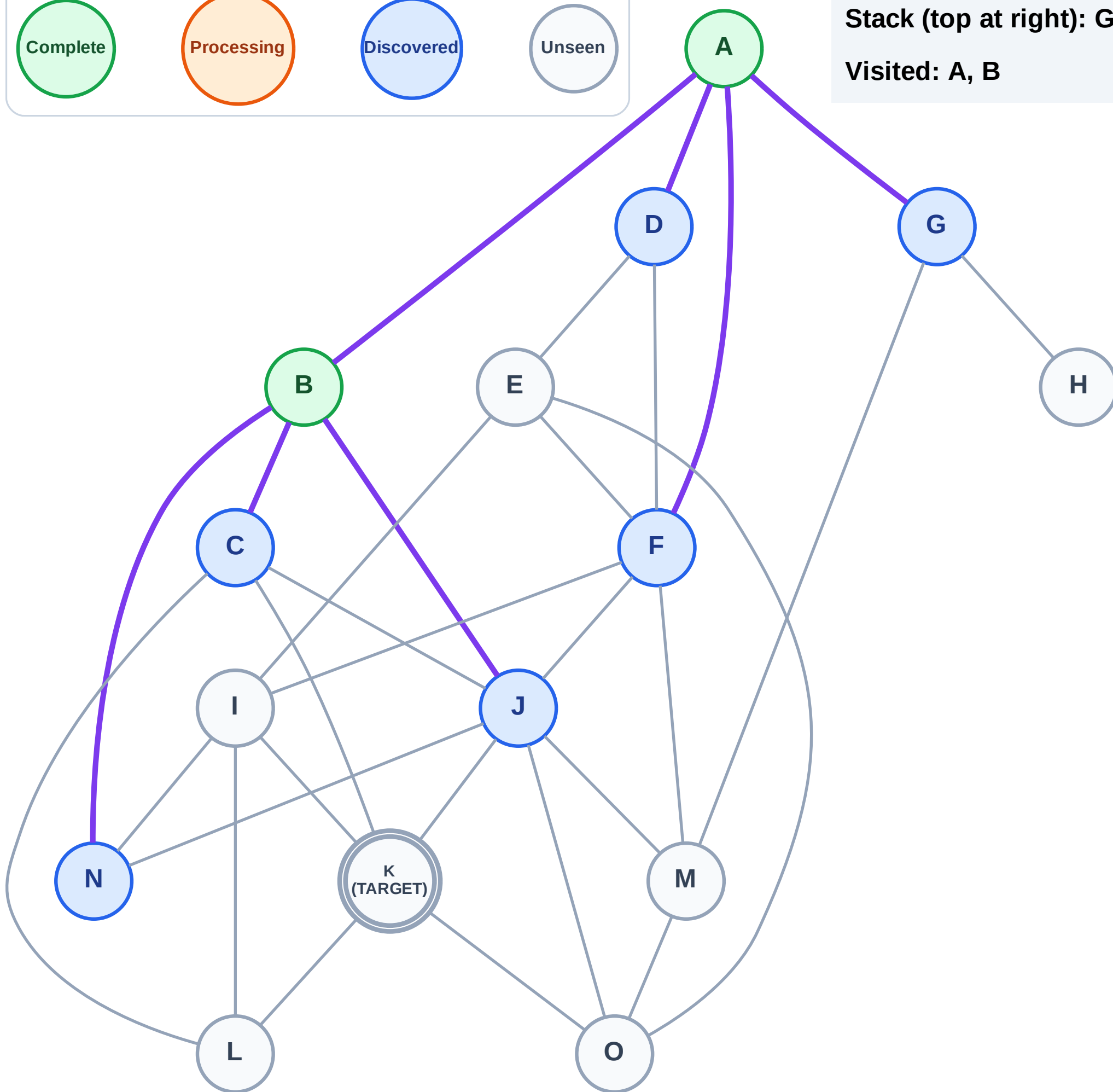
Step 5: Discover N, J, C from B

Legend



Stack (top at right): G | F | D | N | J | C

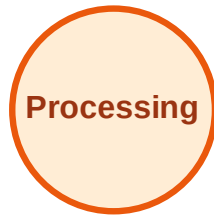
Visited: A, B



DFS Traversal

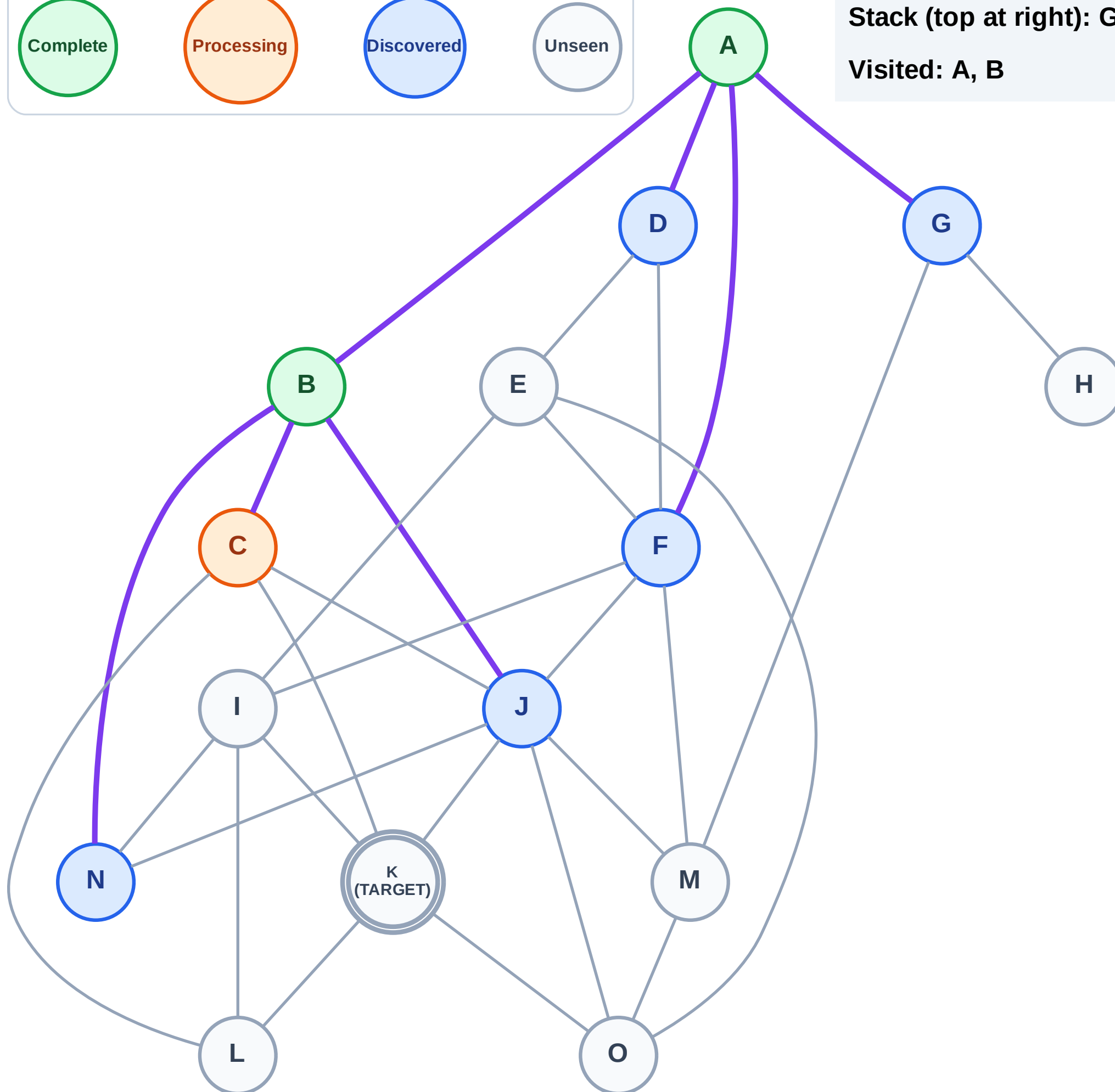
Step 6: Process node C

Legend



Stack (top at right): G | F | D | N | J

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | D | N | J | L
Visited: A, B, C, K

